

THE HOPECORE PROJECT

BCA / Computer Science

Neso Academy

Lang: **EN**

Covers DS, Algorithms, DBMS, CN, OS — structured, exam-focused, brilliant for BCA/BTech CS.

CodeWithHarry

Lang: **HI/EN**

Python, Java, Web Dev, DSA — India's top CS educator. Practical and beginner-friendly.

CS50 Harvard

Lang: **EN**

World's best intro to CS — covers C, Python, Web, SQL, and beyond. University-grade quality.

Abdul Bari

Lang: **EN**

Data Structures & Algorithms explained perfectly. Best on YouTube for DSA theory and logic.

Apna College

Lang: **HI**

DSA in Java, full-stack web — one of the biggest free coding bootcamps for Indian students.

freeCodeCamp

Lang: **EN**

Full courses on Python, Web Dev, ML, Cybersecurity — entirely free and project-based.

Chemical Engineering ★ NEW

LearnChemE (Univ. Colorado Boulder)

Lang: **EN**

Backed by University of Colorado Boulder professors. Covers mass balances, thermodynamics, reaction engineering — the gold standard for ChemE on YouTube.

NPTEL Chemical Engineering

Lang: **EN**

IIT professors on Process Control, Chemical Reaction Engineering, Separation Processes — full university-level courses, completely free.

Chemical Engineering Guy

Lang: **EN**

Process engineering, distillation, furnace design, material balances — practical and industry-oriented. Ideal for ChemE undergrads.

The Organic Chemistry Tutor

Lang: **EN**

Physical chemistry, thermodynamics, fluid mechanics — extremely clear worked examples for university ChemE students.

Khan Academy Chemistry

Lang: **EN**

Organic, inorganic & physical chemistry from scratch — reliable and systematic for ChemE foundation subjects.

Tyler DeWitt

Lang: **EN**

Chemistry concepts made fun and intuitive — stoichiometry, bonding, reactions. Great for Year 1 ChemE students.

Biology (BSc / B.Tech Biotechnology) ★ NEW

Ninja Nerd Science

Lang: **EN**

Biology, anatomy, physiology, biochemistry — detailed whiteboard-style lectures. Huge following among university biology students worldwide.

MIT OCW — Biology

Lang: **EN**

Full MIT lectures — Intro Biology, Genetics, Biochemistry, Cell Biology (7.012/7.013/7.014). Deepest free biology content available.

Amoeba Sisters

Lang: **EN**

Genetics, cell biology, ecology — animated with clarity and humour. Perfect for building solid foundations at undergraduate level.

CrashCourse Biology

Lang: **EN**

50-episode full biology series — cells, evolution, genetics, ecology. Fast, engaging, and genuinely university-level.

Shomu's Biology

Lang: **EN**

Microbiology, Biochemistry, Molecular Biology, Immunology — comprehensive for BSc/MSc biology students.

iBiology

Lang: **EN**

Lectures by world-leading scientists — 15-20 new talks per year. Covers cell biology, genetics, neuroscience at research depth.

Medicosis Perfectionalis

Lang: **EN**

Microbiology, pathology, pharmacology — highly structured and detailed. Top pick for biomedical science students.

Bozeman Science

Lang: **EN**

AP/university biology — genetics, evolution, biochemistry — by award-winning teacher Paul Andersen. Clear and trustworthy.

Engineering — Core Subjects

NPTEL / IIT Professors

Lang: **EN**

IIT & IISc professors teaching every engineering subject. India's most reliable university-grade source.

The Engineering Mindset

Lang: **EN**

HVAC, Electrical, Fluid Mechanics — clear diagrams and real-world examples. Excellent foundation.

Learn Engineering

Lang: **EN**

3D animated explanations of engines, machines, thermodynamics. Perfect for visual learners.

Neso Academy

Lang: **EN**

Digital Electronics, Signals & Systems, Engineering Maths — essential for ECE/EEE students.

Veritasium	Lang: EN
Deep science & engineering concepts via demos and experiments. Builds real understanding.	
Engineer4Free	Lang: EN
500+ videos on structural analysis, mechanics of materials, project management. Very reliable.	

Mechanical Engineering

Learn Engineering	Lang: EN
How engines, turbines, gearboxes work — stunning 3D animations, covers all ME subjects.	
Engineering Explained	Lang: EN
Thermodynamics, fluid mechanics, automotive engineering explained simply and accurately.	
NPTEL Mechanical	Lang: EN
IIT professors on Heat Transfer, Manufacturing, Machine Design — full university-level courses.	
SmarterEveryDay	Lang: EN
Real-world engineering physics — aerospace, fluid dynamics, high-speed camera experiments.	
The Efficient Engineer	Lang: EN
Stress analysis, FEA, thermodynamics — concise and beautifully presented for ME students.	

Civil Engineering

Engineer4Free	Lang: EN
Structural analysis, statics, mechanics of materials — detailed and exam-ready tutorials.	
NPTEL Civil	Lang: EN
IIT faculty on Geotechnical, Fluid Mechanics, Structural Engineering — full semester courses.	
Practical Engineering	Lang: EN
Bridges, dams, infrastructure — explains how civil structures actually work in the real world.	
The B1M	Lang: EN
World's biggest construction channel — megaprojects, building tech, and industry insights.	

Electronics Engineering (ECE/EEE)

Neso Academy	Lang: EN
Digital Electronics, Microprocessors, Signals & Systems — best structured ECE channel on YouTube.	
All About Electronics	Lang: EN

Op-amps, transistors, filters — practical circuit theory explained clearly for ECE students.	
ElectroBOOM	Lang: EN
Electronics fundamentals with humour — circuit theory, power supplies, AC/DC. Highly engaging.	
NPTEL ECE	Lang: EN
IIT professors on VLSI, Communication Systems, Analog Circuits — full university courses.	

Data Science, AI & Machine Learning

StatQuest — Josh Starmer	Lang: EN
Statistics, ML algorithms, deep learning — visualised brilliantly. Best for building real understanding.	
3Blue1Brown	Lang: EN
Linear algebra, calculus, neural networks — the most beautiful math visualisations on YouTube.	
Andrej Karpathy	Lang: EN
Deep learning, LLMs from scratch — former Tesla AI Director. Engineer-grade explanations.	
Sentdex	Lang: EN
Python, ML, TensorFlow — practical tutorials. Great for building real ML projects from scratch.	
Two Minute Papers	Lang: EN
Latest AI research summarised in 2 mins. Perfect for staying current on breakthroughs.	
codebasics	Lang: EN/HI
Data Analytics, Data Science projects, career advice — industry-focused practical learning.	

Business / BBA / MBA

Harvard Business Review	Lang: EN
Strategy, leadership, case studies from the world's top business school. Essential for MBA.	
Wharton School	Lang: EN
Finance, marketing, operations — Ivy League university lectures accessible completely free.	
Khan Academy Economics	Lang: EN
Micro & Macroeconomics from scratch — supply-demand, GDP, monetary policy, perfectly explained.	
CrashCourse Business	Lang: EN
Entrepreneurship, accounting, marketing — fun, fast, and genuinely university-level content.	
Aswath Damodaran (NYU)	Lang: EN

Corporate Finance & Valuation by NYU Stern professor — considered the best finance teacher alive.

The Economist

Lang: **EN**

Global economics, policy, markets — essential watching for MBA students and business graduates.

Finance & Accounting

Aswath Damodaran (NYU)

Lang: **EN**

Valuation, corporate finance, investment strategy — NYU professor, best finance educator on YouTube.

CA Wallah (Physics Wallah)

Lang: **HI/EN**

CA Foundation, Intermediate — ICAI-aligned, extremely popular for Indian commerce students.

Accounting Stuff

Lang: **EN**

Balance sheets, P&L, financial statements — explained with excellent visuals for beginners.

The Plain Bagel

Lang: **EN**

Investing, bonds, portfolio theory, risk — real finance concepts demystified at university level.

University Mathematics

3Blue1Brown

Lang: **EN**

Calculus, Linear Algebra, Probability — visualised unlike anywhere else. Changes how you see maths.

MIT OpenCourseWare

Lang: **EN**

Full courses — Calculus, Linear Algebra, Differential Equations — taught by MIT professors.

Professor Leonard

Lang: **EN**

Calculus 1, 2, 3 — long, thorough lecture-style videos. Rated the best Calculus teacher on YouTube.

PatrickJMT

Lang: **EN**

Engineering Maths, Statistics — thousands of worked examples. Excellent for exam preparation.

University Physics

MIT OpenCourseWare

Lang: **EN**

University Physics, Quantum Mechanics, Thermodynamics — full MIT professor lectures for free.

Physics with Professor Dave

Lang: **EN**

Classical Mechanics, Electromagnetism, Modern Physics — concise, clear, university-level.

Veritasium

Lang: **EN**

Quantum physics, relativity, entropy — deep dives into the hardest concepts as video essays.

DrPhysicsALang: **EN**

Relativity, Quantum Mechanics, Nuclear Physics — UK physicist with clear whiteboard teaching.

General / All University Degrees**MIT OpenCourseWare**Lang: **EN**

Free full courses from MIT — every engineering, science, and business subject. Unmatched quality.

Khan AcademyLang: **EN**

University maths, economics, computing — structured, free, and the world's most trusted educator.

CrashCourseLang: **EN**

Covers every university subject — history, economics, biology, CS — fast, fun, accurate.

NPTELLang: **EN**

India's national platform — IIT/IISc professors teaching every UG/PG subject completely for free.

TED-EdLang: **EN**

Short animated university-level lessons on science, philosophy, economics — brilliant for depth.

Chemical Eng: LearnChemE + NPTEL | Biology: Ninja Nerd + MIT OCW + Shomu's Biology | All subjects: MIT OCW + NPTEL

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